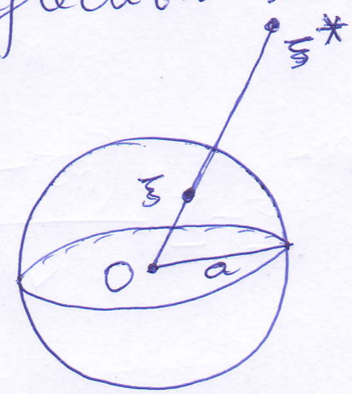
Dirichlet problem on a Ball

We again use the method of reflection to construct the Green's function.

Let $\Omega \equiv \{ x \in \mathbb{R}^n : |x| < a \}$.

For $\xi \in \Omega$, define $\xi^* = \dfrac{a^2 \xi}{|\xi|^2}$

as its reflection in $\partial\Omega$.

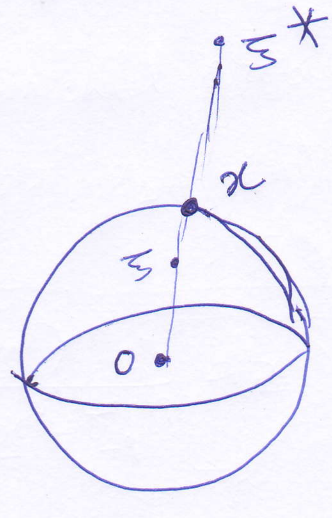


$$|\xi| < a$$

$$\frac{1}{|\xi|} > \frac{1}{a}$$

$$|\xi^*| = \frac{a^2 |\xi|}{|\xi|^2} = \frac{a^2}{|\xi|} > \frac{a^2}{a} = a.$$

$$\Rightarrow \xi^* \notin \Omega.$$



For $|x| = a$

$$|x - \xi^*|^2 = \left(x - \frac{a^2 \xi}{|\xi|^2} \right)^2$$

(2)

$$= x^2 - 2\frac{a^2}{|\xi|^2} x\cdot\xi + \frac{a^4\xi^2}{|\xi|^4}$$

$$= a^2 - 2\frac{a^2}{|\xi|^2}\sum_{i=1}^{n} x_i\xi_i + \frac{a^4}{|\xi|^2}$$

$$= \frac{a^2}{|\xi|^2}\left(a^2 - 2\sum_{i=1}^{n} x_i\xi_i + |\xi|^2\right)$$

$$= \frac{a^2}{|\xi|^2}|x-\xi|^2.$$

Thus we obtain,

$$\frac{|x-\xi^*|}{|x-\xi|} = \frac{a}{|\xi|} \quad \text{for } |x| = a.$$

$$\Longrightarrow |x-\xi| = \frac{|\xi|}{a}|x-\xi^*|$$

So

$$K(x-\xi) = \begin{cases} \frac{1}{2\pi}\log\left(\frac{|\xi|}{a}|x-\xi^*|\right), & n=2. \\ \left(\frac{a}{|\xi|}\right)^{n-2} K(x-\xi^*), & n\geq 3. \end{cases}$$

Define for $x, \xi \in \Omega$ ③.

$$G(x,\xi) = \begin{cases} K(x-\xi) - \frac{1}{2\pi}\log\left(\frac{|\xi|}{a}|x-\xi^*|\right), & n=2 \\ K(x-\xi) - \left(\frac{a}{|\xi|}\right)^{n-2} K(x-\xi^*), & n\geq 3 \end{cases}$$

Since ξ^* is not in Ω, the second terms in the RHS of $G(x,\xi)$ are harmonic in ~~x~~ $\in \Omega$.

Moreover we have $G(x,\xi) = 0$ if $x \in \partial\Omega$.

Thus G defined in the above is the Greens function for Ω.

Now let us calculate the Poisson Kernel on $\partial\Omega$. We first compute the gradient of $G(x,\xi)$ for $n \geq 2$.

$$\frac{\partial}{\partial x_i} G(x,\xi) = \frac{(x_i - \xi_i)}{\omega_n |x-\xi|^n} - \frac{(x_i - \xi_i^*)}{\omega_n |x-\xi^*|^n}\left(\frac{a}{|\xi|}\right)^{n-2}$$

Using $\dfrac{1}{|x-\xi^*|^n} = \dfrac{|\xi|^n}{a^n}|x-\xi|^{-n}$

we have $\dfrac{\partial G(x,\xi)}{\partial x_i}$

$$= \frac{1}{|x-\xi|^n}\frac{1}{\omega_n}\left[(x_i-\xi_i) - \left(x_i - \frac{\xi_i a^2}{|\xi|^2}\right)\frac{|\xi|^2}{a^2}\right]$$

$$= \frac{1}{\omega_n |x-\xi|^n} \cdot x_i \left(1 - \frac{|\xi|^2}{a^2}\right)$$

Because $x \in \partial\Omega$, $|x| = a$.

Outward unit normal vector

$\hat{n} = \dfrac{\nabla f}{|\nabla f|}$ where $f(x)$ is the surface equation

Here, the surface is $n-1$ dimensional sphere.

$$f(\underline{x}) = x_1^2 + x_2^2 + \cdots + x_n^2 - a^2$$

$$\nabla f(\underline{x}) = (2x_1, 2x_2, \ldots, 2x_n)$$

$$|\nabla f| = 2\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$
$$= 2|x|$$
$$= 2a.$$

$$\hat{n} = \frac{1}{a}(x_1, x_2, \ldots, x_n) = \frac{x}{a}.$$

So the Poisson Kernel is

$$H(x, \xi) = \nabla G \cdot \hat{n}$$
$$= \frac{a^2 - |\xi|^2}{a\,\omega_n |x - \xi|^n}, \text{ for } x \in \partial\Omega,$$

Now the solution of the Dirichlet problem when $\Omega \equiv B_a(0)$ is given by the Poisson integral formula:

$$u(\xi) = \frac{a^2 - |\xi|^2}{a\omega_n} \int_{|x|=a} \frac{g(x)}{|x-\xi|^n}\, dS_x.$$

where $g(x)$ is the boundary data:

$$u(x) = g(x) \text{ on } \partial\Omega.$$